

**Practical File
Of
Computer Network Laboratory
(UGCA1916)**



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Computer Network Practical File Index Format

Sr. No.	Name of the Experiment/Program	Date of Submission	Page No.	Teacher's Signature	Remarks
1	Familiarization with networking components and devices: LAN adapters, Hubs, Switches, Routers etc.		3		
2	Familiarization with transmission media tools: Coaxial cable, UTP cable, Crimping tool, Connectors etc.		5		
3	Preparing straight and cross etc.		7		
4	Study of various LAN topologies and their creation using network devices, cables, and computers.		10		
5	Configuration of TCP/IP Protocols in Window and Linux.		12		
6	Implementation of resource sharing (file, printer etc.)		14		
7	Designing and implementation class A, B, and C networks.		16		
8	Subnet planning and its implementation.		18		
9	To configure dynamic IP address for a computer connected to a LAN.		20		
10	Use of commands like ping, ipconfig for trouble shooting network related problems.		22		
11	Develop a program to compute the Hamming Distance between any two code words.		24		
12	Installation of FTP server and client.		25		
13	To configure proxy server.		27		
14	Familiarization with network simulation tools.		29		

1. Familiarization with networking components and devices: LAN adapters, Hubs, Switches, Routers etc.

Components & Theory

1. LAN Adapter (NIC)

- Connects computer to LAN.
- Converts data into signals.
- Types: Wired (Ethernet), Wireless (Wi-Fi).

Your computer's passport to the network. It converts digital data into signals for transmission and vice-versa. Think of it as the translator for your PC.

2. Hub

- Works at **Physical Layer (Layer 1)**.
- Connects devices, broadcasts data to all.
- Simple, causes collisions.

A simple, Layer 1 device that connects multiple Ethernet devices. When it receives data, it broadcasts it to all connected devices, which can lead to network collisions.

3. Switch

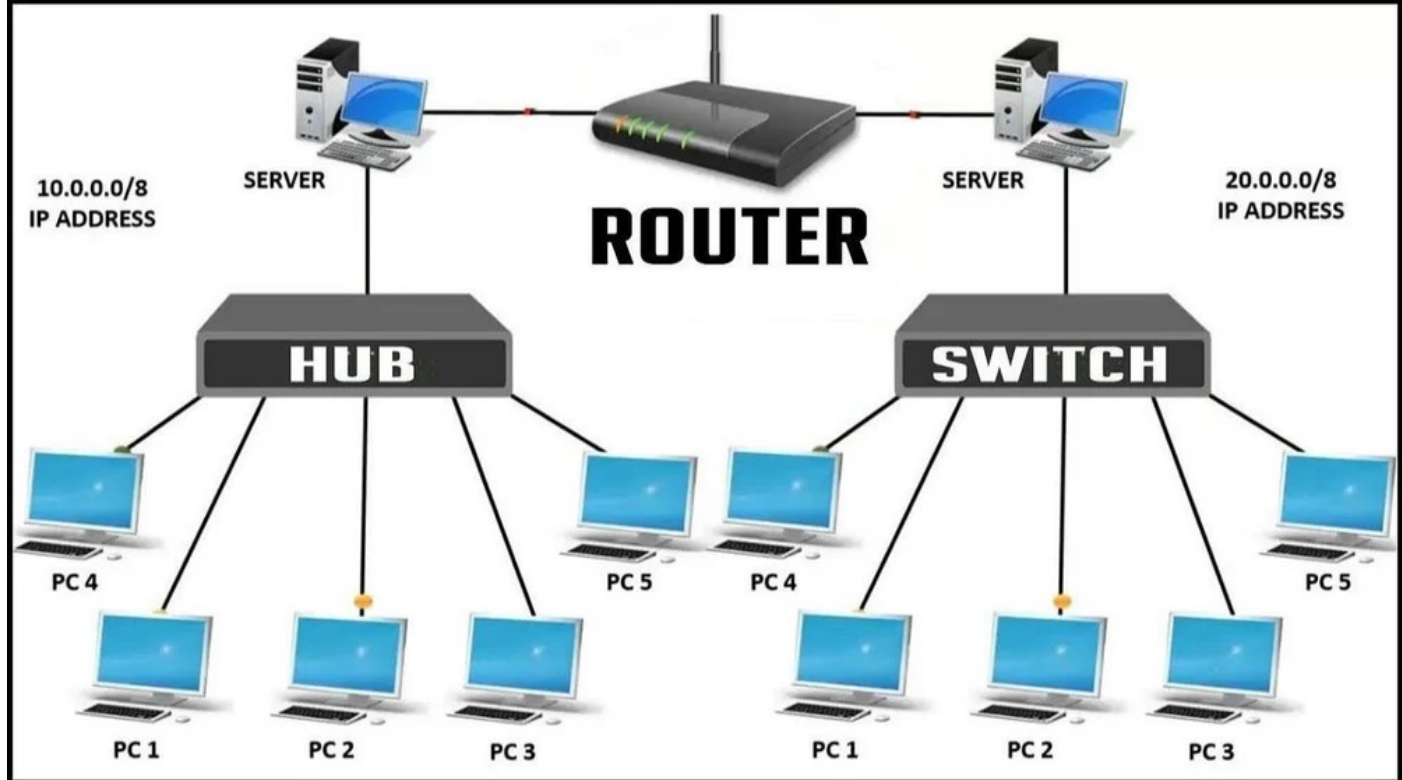
- Works at **Data Link Layer (Layer 2)**.
- Uses **MAC address** to send data only to target device.
- Efficient, reduces collisions.

Smarter than a hub, this Layer 2 device uses MAC addresses to send data only to the intended recipient, improving efficiency and reducing collisions.

4. Router

- Works at **Network Layer (Layer 3)**.
- Connects different networks (LAN ↔ Internet).
- Uses **IP address**, supports DHCP, NAT, Firewall.

The traffic cop of your network! It connects different networks (like your home LAN to the internet) and uses IP addresses to direct data packets to their destinations.



Conclusion

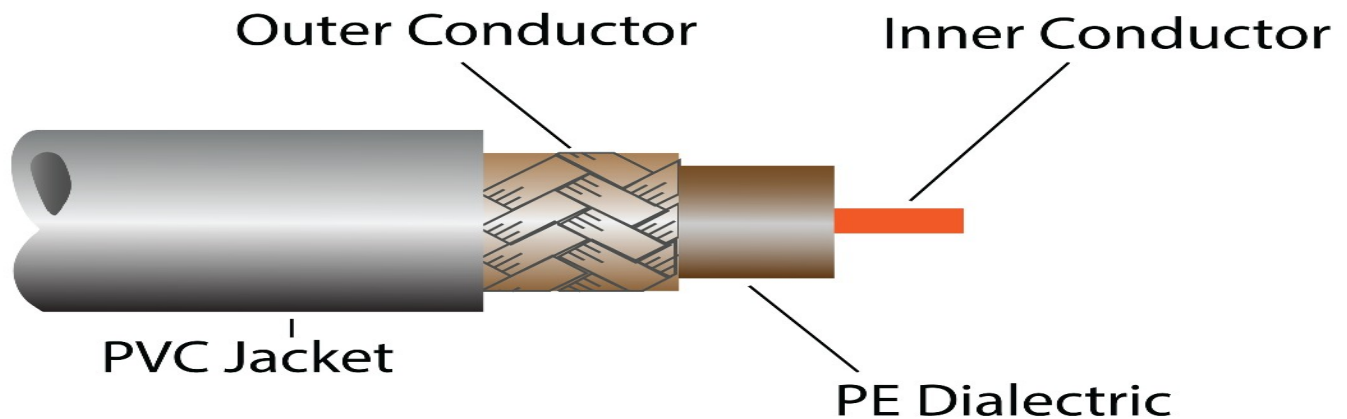
- LAN Adapter: connects PC to network.
- Hub: broadcasts to all devices.
- Switch: forwards to correct device.
- Router: connects networks, provides internet.

2. Familiarization with transmission media tools: Coaxial cable, UTP cable, Crimping tool, Connectors etc.

Networking isn't just about the devices; it's also about how data travels between them. This section explores the physical cables and tools used to build these connections.

Components & Theory

The Coaxial Cable was once common for TV and older LANs. Its design features a central copper core, surrounded by an insulating layer, and then a metallic shield. This structure provides good protection against electromagnetic interference.



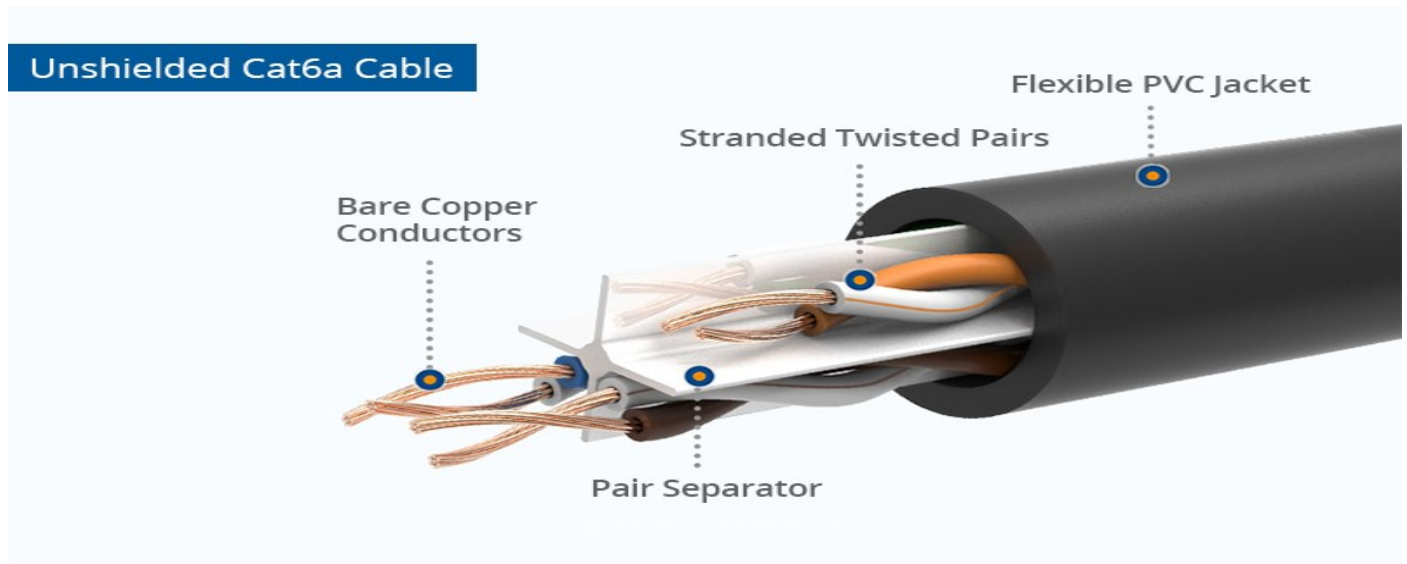
1. UTP Cable (Unshielded Twisted Pair)

Unshielded Twisted Pair (UTP) cables are the most common type of networking media used in modern **Ethernet Local Area Networks (LANs)**. They are highly popular due to their flexibility, ease of installation, and cost-effectiveness.

Key Characteristics

- **Structure:** Consists of **twisted pairs of copper wires** encased in a plastic outer jacket, with **no metallic shielding** (hence "Unshielded").
 - *Note:* The twisting of the wires helps minimize **crosstalk** (interference between adjacent wire pairs) and external **Electromagnetic Interference (EMI)**.
- **Application:** Primarily used for data and voice transmission in **Ethernet LANs** (e.g., connecting computers to switches or routers).
- **Cost & Installation:** They are generally the **cheapest** cabling option and are **flexible and easy to install** compared to shielded or fiber optic cables.
- **Categories (Performance Grades):** UTP cables are standardized into different categories based on their performance and maximum data rates:
 - **Cat5e:** Supports up to 1 Gigabit per second (Gbps).
 - **Cat6:** Supports 1 Gbps up to 100 meters, and 10 Gbps for shorter distances (up to 55 meters).

- **Cat6A:** Supports 10 Gbps up to 100 meters.

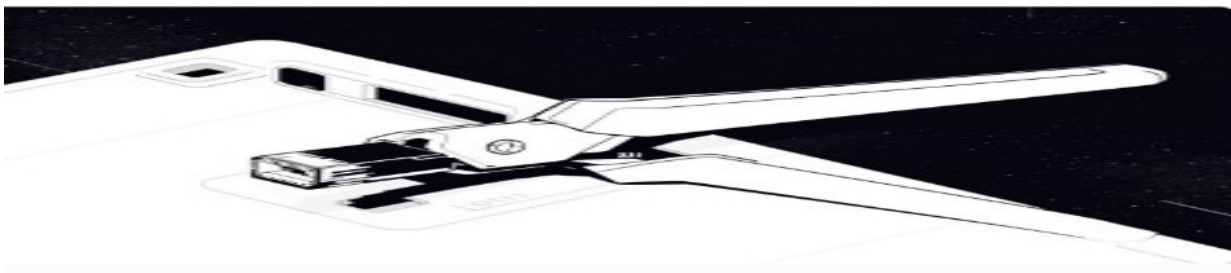


2. Crimping Tool

The **Crimping Tool** is an **essential hand tool** used in networking to **securely attach connectors** (such as **RJ-45** or **RJ-11**) to the ends of network cables (like UTP).

Key Functions

- **Attachment:** Physically locks the plastic connector housing (e.g., RJ-45) onto the cable jacket.
- **Contact:** Simultaneously presses the metal pins within the connector down, forcing them to pierce the insulating jackets and make **proper electrical contact** with the copper wires inside the cable.
- **Cable Customization:** It is necessary for technicians who need to create custom-length patch cables or terminate cables for wall jacks and panels. **A successful crimp ensures reliable data transmission.**



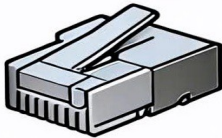
3. Connectors

Network Connectors are physical interface devices used to terminate the ends of networking cables and **join them to network equipment** (such as switches, routers, or computers). They are essential for creating a reliable and secure physical connection, which allows for signal transmission.

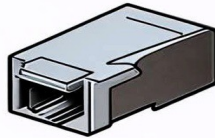
Key Functions and Examples

- **Function:** To provide a stable and secure physical and electrical interface between the cable media and the active networking device port.
- **Physical Connection:** They ensure the cable's internal conductors (e.g., copper wires or fiber strands) align correctly with the device port contacts.
- **Common Examples:**
 - **RJ-45 (Registered Jack 45):** The standard connector used to terminate **Unshielded Twisted Pair (UTP)** and **Shielded Twisted Pair (STP)** Ethernet cables.
 - **BNC (Bayonet Neill–Concelman):** Used for **coaxial cables**, historically common in older networks (like 10BASE2) and still used for video signals.
 - **Fiber Connectors (e.g., LC, SC):** Used for terminating **Fiber Optic Cables**; they rely on precision alignment to transmit light signals.

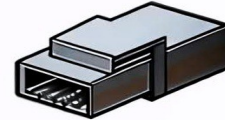
ETHERNET CONNECTORS



RJ45

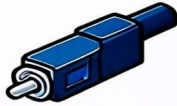


GG45

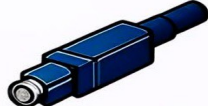


Tera

FIBER OPTICS CONNECTORS



SC



LC



ST



MTP/MPO

COAXIAL CABLE CONNECTORS



BNC



F-Type



N-Type



N-Type

TELEPHONE & SERIAL CONNECTORS



RJ11



DB9 (RS-232)

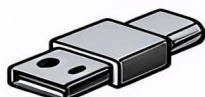


DB25

POWER & MISC. CONNECTORS



DC Connectors



USB G+



USB or C



Keystone Jacks

Conclusion

We studied transmission media tools:

- **Coaxial cable:** shielded, used for TV/older LAN.
- **UTP cable:** common in Ethernet LANs.
- **Crimping tool:** attaches connectors to cables.
- **Connectors:** provide secure cable-to-device connection.

3. Preparing straight and cross etc.

Theory

1. Straight Cable (Ethernet Straight-Through)

- Used to connect **different types of devices** (e.g., PC → Switch, PC → Router).
- Both ends of the cable have the **same wiring sequence** (T568A–T568A or T568B–T568B).
- Common in LAN setups for connecting computers to networking devices.

2. Cross Cable (Ethernet Crossover)

- Used to connect **similar devices directly** (e.g., PC → PC, Switch → Switch).
- One end uses **T568A** wiring and the other end uses **T568B** wiring.
- Swaps transmit and receive lines to enable direct communication.

Wiring Standards (Color Codes)

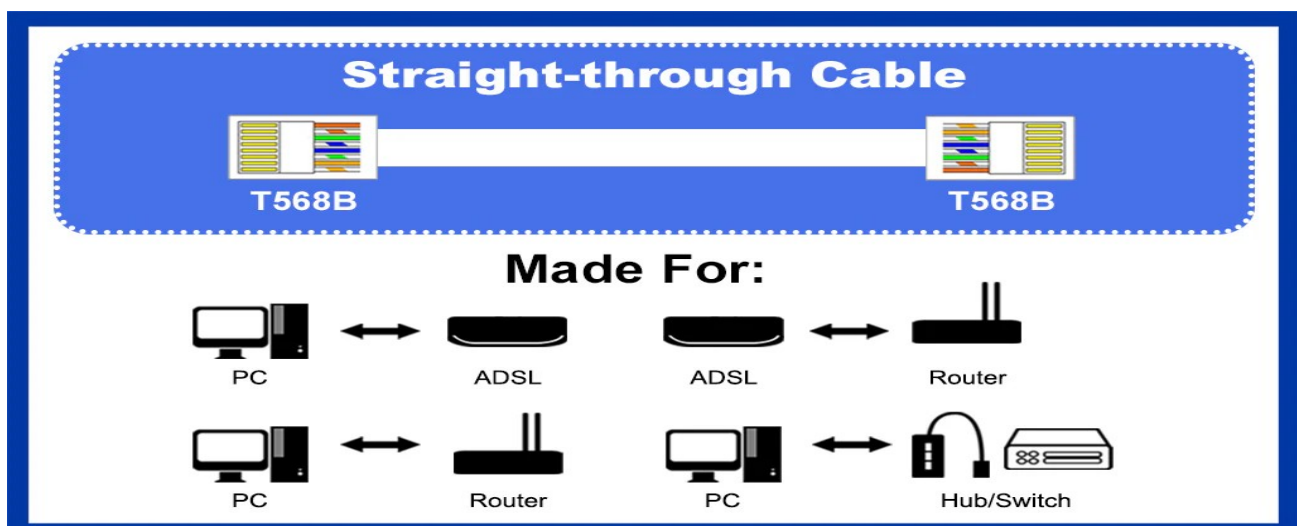
- **T568A Sequence:** Green/White, Green, Orange/White, Blue, Blue/White, Orange, Brown/White, Brown
- **T568B Sequence:** Orange/White, Orange, Green/White, Blue, Blue/White, Green, Brown/White, Brown

Procedure (Practical Steps)

1. Cut UTP cable to required length.
2. Strip outer jacket carefully.
3. Arrange wires in required sequence (T568A or T568B).
4. Insert wires into RJ-45 connector.
5. Use **crimping tool** to fix connector.
6. Repeat for other end according to straight or cross type.

Conclusion

- **Straight cable** connects different devices (PC → Switch/Router).
- **Cross cable** connects similar devices (PC → PC, Switch → Switch).
- Proper wiring standards (T568A/B) and crimping ensure successful communication.



4. Study of various LAN topologies and their creation using network devices, cables, and computers.

Theory

1. Bus Topology

- Single backbone cable connects all computers.
- Simple, low cost, but collisions occur.
- Example: Coaxial cable networks.

2. Star Topology

- All computers connect to a central device (Hub/Switch).
- Easy to install, easy fault detection.
- Most common in modern LANs.

3. Ring Topology

- Each computer connects to two others, forming a ring.
- Data travels in one direction.
- Failure of one node can affect the whole network.

4. Mesh Topology

- Every device connects to every other device.
- Provides redundancy and reliability.
- Expensive, used in critical networks.

5. Hybrid Topology

- Combination of two or more topologies (e.g., Star + Bus).
- Flexible, scalable, widely used in large organizations.

Creation Using Devices

- **Bus:** Coaxial cable + terminators.
- **Star:** UTP cable + Hub/Switch + PCs.
- **Ring:** UTP/Coaxial cables + computers arranged in ring.
- **Mesh:** Multiple UTP cables + routers/switches.
- **Hybrid:** Mix of above devices and cables.

Conclusion

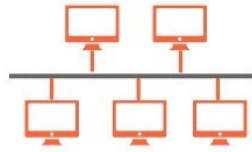
We studied LAN topologies:

- **Bus:** simple backbone, collisions.
- **Star:** central device, most common.
- **Ring:** circular path, one failure affects all.
- **Mesh:** full connectivity, reliable but costly.
- **Hybrid:** combination, flexible for large networks.

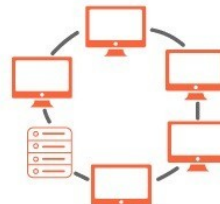
Network Topology Types



Point to Point



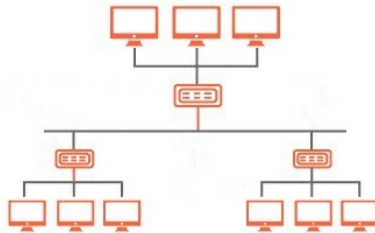
Bus



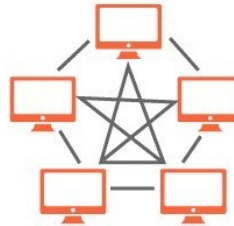
Ring



Star



Tree



Mesh



Line

5. Configuration of TCP/IP Protocols in Window and Linux.

Theory

- **TCP/IP Protocol** is the fundamental suite used for communication in computer networks.
- Configuration involves assigning **IP address, Subnet Mask, Default Gateway, and DNS** to a computer.

Steps in Windows

1. Go to **Control Panel** → **Network and Sharing Center** → **Change adapter settings**.
2. Right-click on **Local Area Connection / Wi-Fi** → **Properties**.
3. Select **Internet Protocol Version 4 (TCP/IPv4)** → Click **Properties**.
4. Enter:
 - IP Address (e.g., 192.168.1.10)
 - Subnet Mask (e.g., 255.255.255.0)
 - Default Gateway (e.g., 192.168.1.1)
 - Preferred DNS (e.g., 8.8.8.8)
5. Save settings → Test with ping command.

Steps in Linux

1. Open **Terminal**.
2. To view current settings:

```
ifconfig
```

or

```
ip addr show
```

3. To assign IP temporarily:

```
sudo ifconfig eth0 192.168.1.20 netmask 255.255.255.0
```

or

```
sudo ip addr add 192.168.1.20/24 dev eth0
```

4. To set Default Gateway:

```
sudo route add default gw 192.168.1.1
```

or

```
sudo ip route add default via 192.168.1.1
```

5. For permanent settings, edit config files:

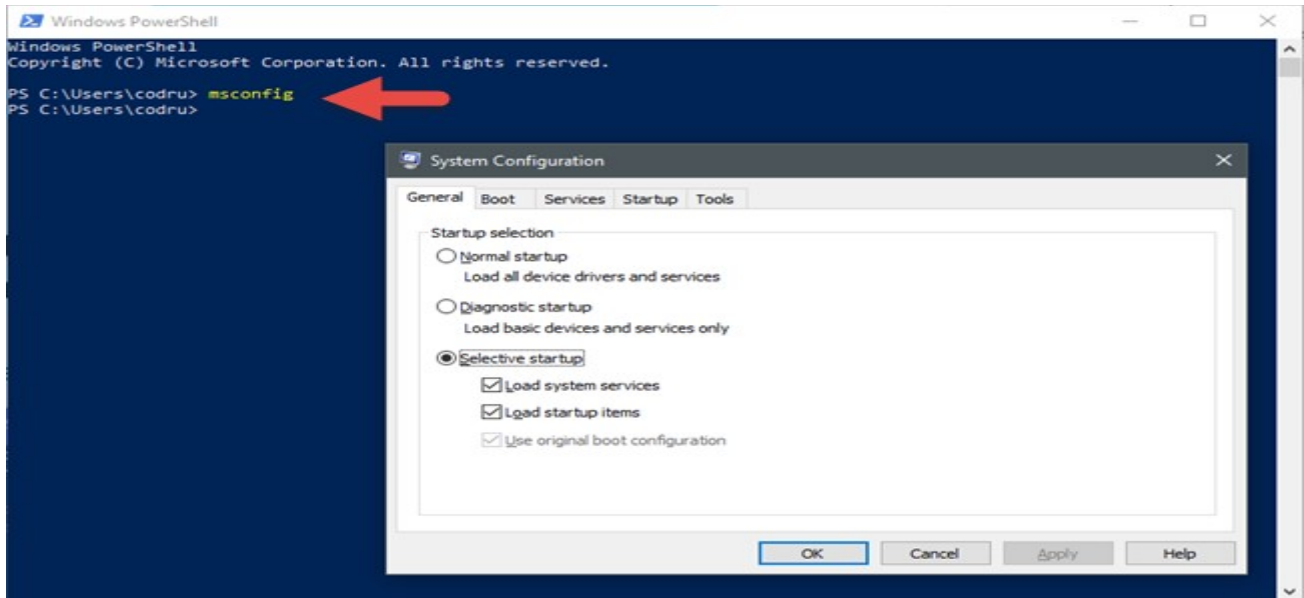
- **Ubuntu/Debian:** /etc/network/interfaces or Netplan YAML files.
- **RedHat/CentOS:** /etc/sysconfig/network-scripts/ifcfg-eth0.

6. Test with ping command.

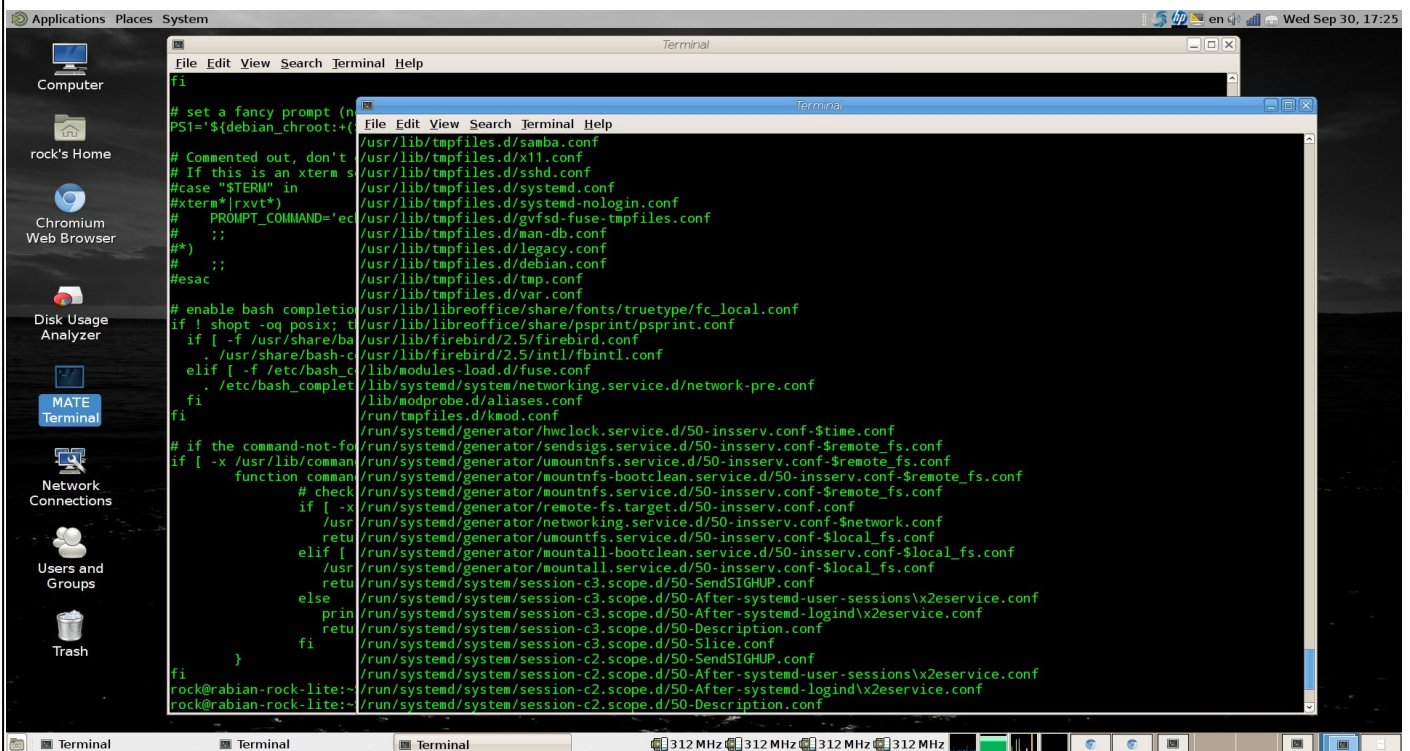
Conclusion

We configured TCP/IP protocols in both Windows and Linux:

- In **Windows**, settings are applied via GUI (Network Properties).
- In **Linux**, configuration is done using **terminal commands** or editing config files.
- Correct TCP/IP setup ensures communication within LAN and Internet access.



Linux Configuration (Termina)



6. Implementation of resource sharing (file, printer etc.)

Theory

- **Resource Sharing** allows multiple computers in a network to access common resources such as files, folders, and printers.
- It improves efficiency, reduces cost, and enables collaboration in LAN environments.

Steps for File Sharing

In Windows:

1. Right-click on a folder → Select **Properties** → **Sharing tab**.
2. Click **Share** → Choose users or “Everyone”.
3. Set permissions (Read/Write).
4. Access shared folder from another PC using \\ComputerName\SharedFolder.

In Linux:

1. Use **Samba service** for sharing files.
2. Edit /etc/samba/smb.conf to define shared folder.
3. Restart Samba service.
4. Access shared folder using network path.

Steps for Printer Sharing

In Windows:

1. Go to **Control Panel** → **Devices and Printers**.
2. Right-click printer → **Printer Properties** → **Sharing tab**.
3. Enable **Share this printer**.
4. Other PCs can add printer via **Network Printer option**.

In Linux:

1. Use **CUPS (Common Unix Printing System)**.
2. Enable printer sharing in CUPS configuration.
3. Other systems can connect via network printer settings.

Conclusion

We implemented resource sharing in LAN:

- **File Sharing** allows multiple users to access common folders.
- **Printer Sharing** enables multiple computers to use a single printer.
- Resource sharing reduces cost and increases collaboration in a network.



7. Designing and implementation class A, B, and C networks.

Theory

- IPv4 addresses were divided into **5 classes**: A, B, C, D, E.
- The class is decided by the **first octet** (first number of the IP address).
- Each class has a **default subnet mask**, which tells the Network part and Host part.
- Purpose:
 - **Class A** → Very large networks
 - **Class B** → Medium networks
 - **Class C** → Small networks

Class Table (Easy)

Classes	First Octet Range	Default Mask	Hosts per Network
A	1–126	255.0.0.0	~16 million
B	128–191	255.255.0.0	~65 thousand
C	192–223	255.255.255.0	254

- **127.x.x.x** → Loopback (not used in networking)
- **0.x.x.x** → Reserved

Implementation (Short Practical Steps)

1. Choose a Lab Topology

- Use a **Star topology** (all PCs connected to a switch).

2. Assign IP Addresses

Class C Example

- Network: **192.168.1.0/24**
- PC1 → 192.168.1.10
- PC2 → 192.168.1.11
- Mask → **255.255.255.0**

Class B Example

- Network: **172.16.0.0/16**
- Mask → **255.255.0.0**

Class A Example

- Network: **10.0.0.0/8**
- Mask → **255.0.0.0**

3. Configure Each Computer

- Enter IP address & subnet mask in Network Settings (or Terminal for Linux).

4. Test Connectivity

- Use ping command between PCs:
ping 192.168.1.11

If ping is successful → network is working.

IP Address Classes

A B C ... Easy as 1 2 3

Class **A** ... First 1 Bit Fixed 0 x x x x x x x . Host . Host . Host

Class **B** ... First 2 Bits Fixed 10 x x x x x x . Network . Host . Host

Class **C** ... First 3 Bits Fixed 110 x x x x x . Network . Network . Host

8. Subnet planning and its implementation.

Theory

- **Subnetting** is the process of dividing a large IP network into smaller sub-networks (subnets).
- It improves network management, security, and efficient use of IP addresses.
- Each subnet has its own **Network ID** and **Host IDs**.

Steps in Subnet Planning

1. **Identify Requirements** – Number of hosts and subnets needed.
2. **Select IP Address Class** – Class A, B, or C depending on network size.
3. **Decide Subnet Mask** – Borrow bits from host portion to create subnets.
 - Example: Class C default mask = 255.255.255.0
 - If 4 subnets needed → borrow 2 bits → new mask = 255.255.255.192
4. **Calculate Subnets and Hosts**
 - Formula:
 - Number of subnets = 2^n (where n = borrowed bits)
 - Hosts per subnet = $2^h - 2$ (where h = host bits left)
5. **Assign IP Ranges** – Allocate IPs to each subnet.

Implementation

In Windows:

1. Go to **Network Adapter Properties** → **TCP/IPv4 Settings**.
2. Enter IP Address, Subnet Mask, Default Gateway.
3. Save and test with ping.

In Linux:

1. Use terminal commands:

```
sudo ifconfig eth0 192.168.1.10 netmask 255.255.255.192
```

or

```
sudo ip addr add 192.168.1.10/26 dev eth0
```

2. Set default gateway:

```
sudo route add default gw 192.168.1.1
```

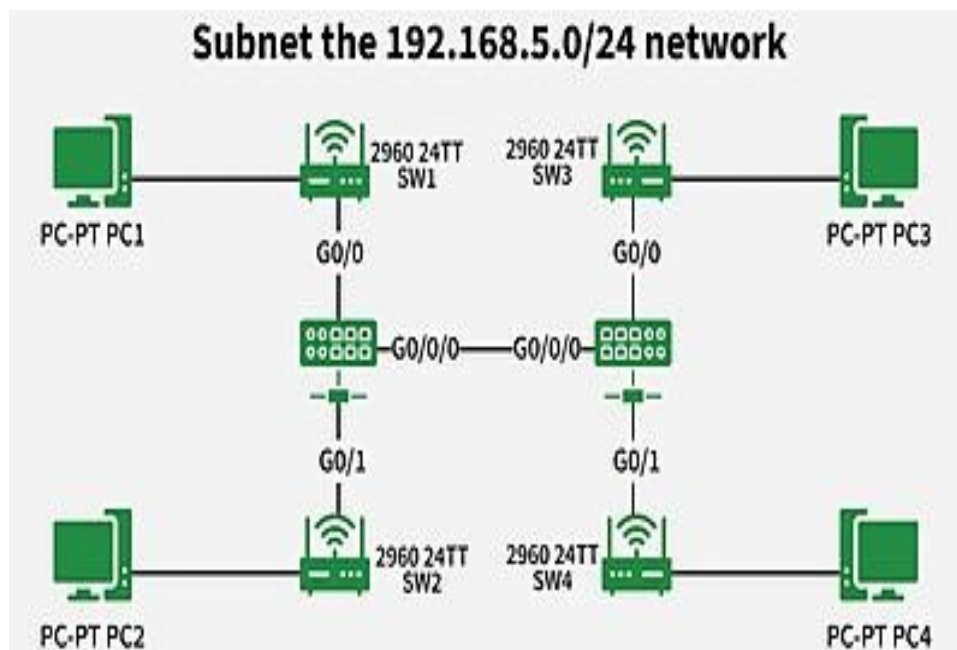
3. Test with ping.

Conclusion

We studied subnet planning and implementation:

- Subnetting divides a large network into smaller subnets.
- Planning involves deciding subnet mask, calculating subnets and hosts, and assigning IP ranges.

- Implementation is done by configuring IP and subnet mask in Windows/Linux.



9. To configure dynamic IP address for a computer connected to a LAN.

Theory

- **Dynamic IP Addressing** means the computer automatically receives an IP address from a **DHCP (Dynamic Host Configuration Protocol) server**.
- DHCP assigns IP, Subnet Mask, Default Gateway, and DNS automatically.
- Advantage: Easy management, no need to manually configure IPs.

Steps in Windows

1. Go to **Control Panel** → **Network and Sharing Center** → **Change adapter settings**.
2. Right-click on **Local Area Connection / Wi-Fi** → **Properties**.
3. Select **Internet Protocol Version 4 (TCP/IPv4)** → Click **Properties**.
4. Choose **Obtain an IP address automatically** and **Obtain DNS server address automatically**.
5. Save settings → Test with ipconfig or ping.

Steps in Linux

1. Open **Terminal**.
2. To enable DHCP temporarily:

```
sudo dhclient eth0
```

3. To check IP:

```
ifconfig
```

or

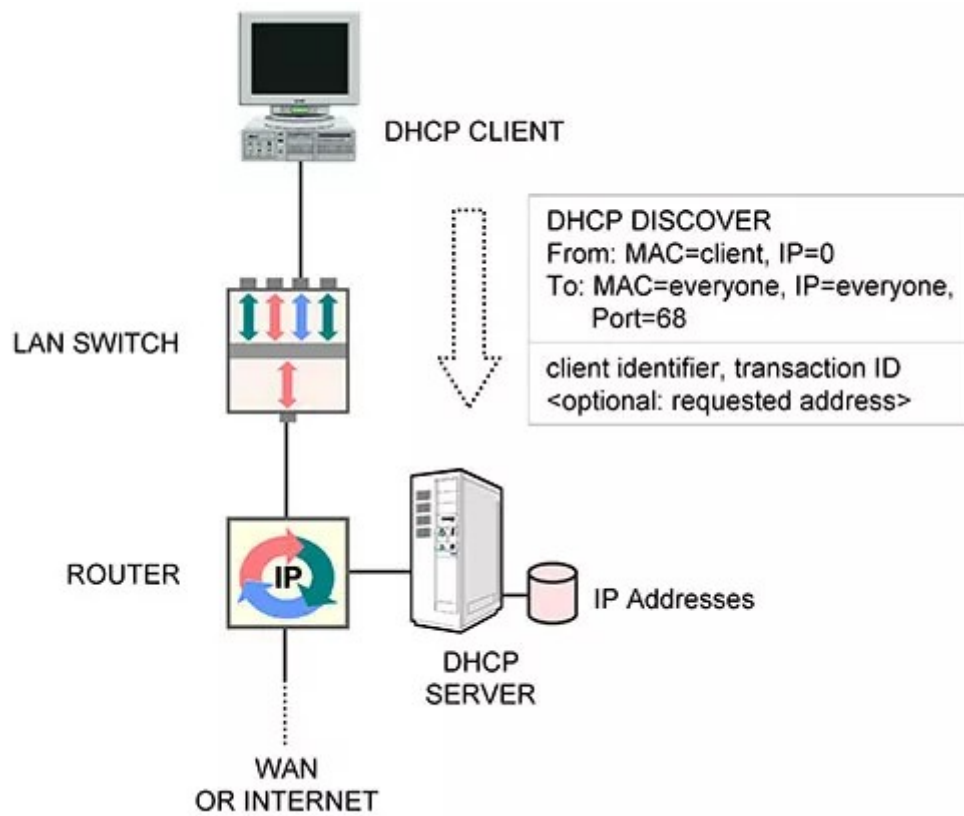
```
ip addr show
```

4. For permanent configuration:
 - Edit network config file (e.g., /etc/network/interfaces or Netplan YAML in Ubuntu).
 - Set interface method to **dhcp**.
 - Restart networking service.

Conclusion

We configured dynamic IP addressing:

- In **Windows**, by selecting automatic IP/DNS in TCP/IPv4 settings.
- In **Linux**, by enabling DHCP using dhclient or editing config files.
- Dynamic IP simplifies network management by automatically assigning addresses.



10. Use of commands like ping, ipconfig for trouble shooting network related problems.

Theory

- Network troubleshooting involves checking connectivity, IP configuration, and communication between devices.
- Common commands:
 - ping → Tests connectivity between two devices.
 - ipconfig (**Windows**) / ifconfig or ip addr (**Linux**) → Displays and manages IP configuration.

Commands and Usage

1. ping Command

- Syntax: ping <IP Address/Hostname>
- Example:

```
ping 192.168.1.1
ping www.google.com
```
- Purpose:
 - Checks if destination is reachable.
 - Measures response time.
 - Detects packet loss.

2. ipconfig (**Windows**)

- Syntax: ipconfig
- Example:

```
ipconfig /all
ipconfig /release
ipconfig /renew
```
- Purpose:
 - Displays IP address, subnet mask, gateway.
 - Refreshes DHCP lease.
 - Helps identify misconfigured IP settings.

3. ifconfig / ip addr (**Linux**)

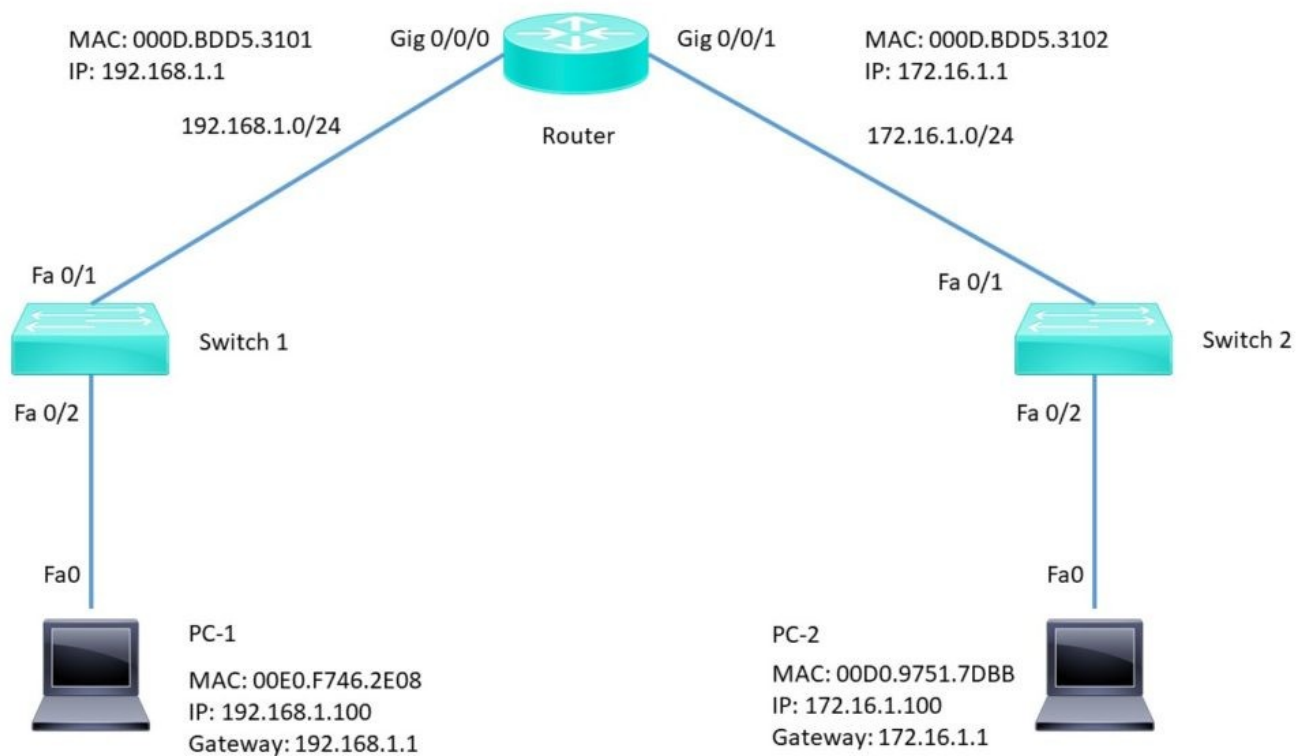
- Syntax: ifconfig or ip addr show
- Example:

```
ifconfig eth0
ip addr show eth0
```
- Purpose:
 - Shows IP configuration of interfaces.
 - Helps verify assigned IP and subnet.

Conclusion

We used commands for troubleshooting:

- ping checks connectivity between devices.
- ipconfig (**Windows**) and ifconfig/ip addr (**Linux**) display and manage IP settings.



11. Develop a program to compute the Hamming Distance between any two code words.

Theory

- **Hamming Distance** is the number of positions at which the corresponding symbols (bits/characters) of two code words are different.
- It is widely used in **error detection and correction** in communication systems.
- Formula:

$dH(x,y)$ = Number of positions where $x_i \neq y_i$ $H(x,y) = \text{Number of positions where } x_i \neq y_i$

Algorithm

1. Input two code words of equal length.
2. Compare each character/bit position.
3. Count the number of mismatches.
4. Output the count as the Hamming Distance.

Program (Python Example)

Program to compute Hamming Distance

```
def hamming_distance(code1, code2):
    if len(code1) != len(code2):
        return "Error: Code words must be of equal length"

    distance = 0
    for i in range(len(code1)):
        if code1[i] != code2[i]:
            distance += 1
    return distance

# Example usage
code1 = "1011001"
code2 = "1001101"

print("Code Word 1:", code1)
print("Code Word 2:", code2)
print("Hamming Distance =", hamming_distance(code1, code2))
```

Sample Output

```
Code Word 1: 1011001
Code Word 2: 1001101
Hamming Distance = 3
```

Conclusion

We developed a program to compute the **Hamming Distance**:

- It counts the number of differing positions between two code words.
- Useful in error detection/correction and coding theory applications.

12. Installation of FTP server and client.

Theory

- **FTP (File Transfer Protocol)** is used to transfer files between computers over a network.
- An **FTP Server** stores files and allows clients to upload/download.
- An **FTP Client** connects to the server to access files.

Steps for FTP Server Installation

In Windows (using FileZilla Server):

1. Download and install **FileZilla Server**.
2. Configure server settings (port, users, passwords).
3. Create shared directories for file storage.
4. Start the server service.

In Linux (using vsftpd):

1. Install package:

```
sudo apt-get install vsftpd
```
2. Edit configuration file `/etc/vsftpd.conf` (enable local users, anonymous access if needed).
3. Restart service:

```
sudo systemctl restart vsftpd
```
4. Add users and set permissions.

Steps for FTP Client Installation

In Windows:

1. Install **FileZilla Client** or use built-in command prompt:

```
ftp <server_ip>
```
2. Enter username and password.
3. Use commands like `get`, `put`, `ls` to transfer files.

In Linux:

1. Install FTP client:

```
sudo apt-get install ftp
```
2. Connect to server:

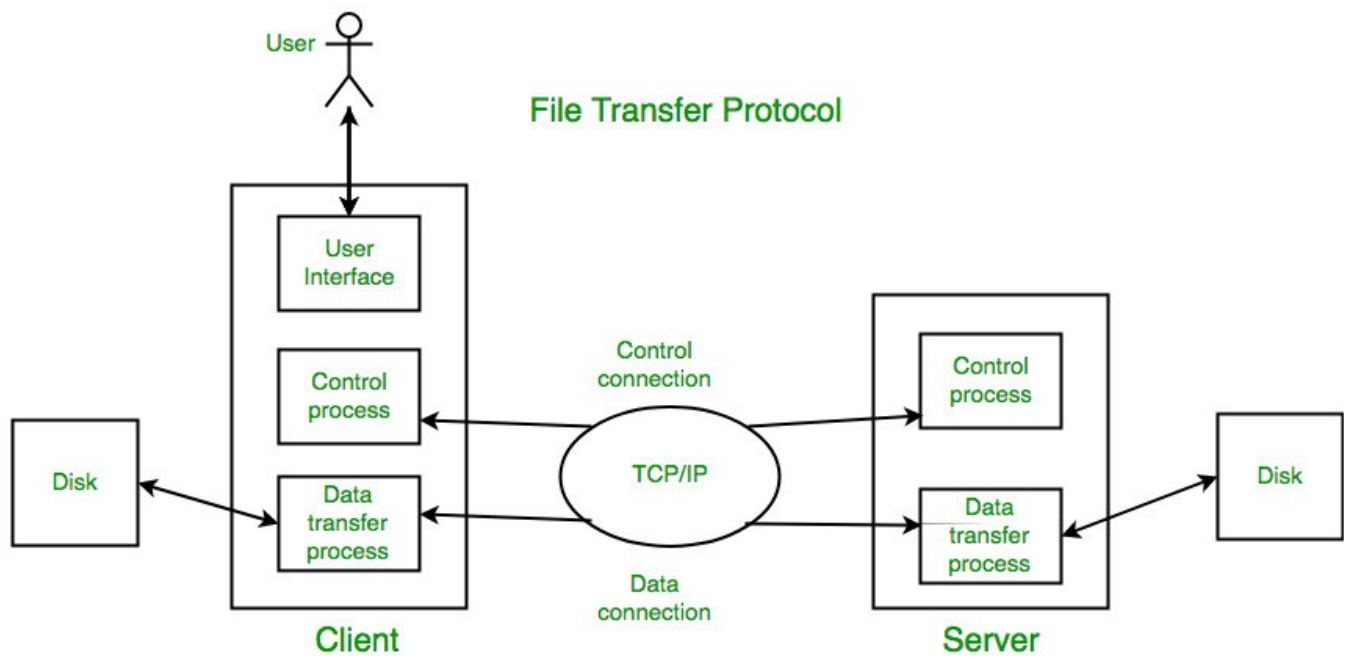
```
ftp <server_ip>
```
3. Authenticate with username and password.
4. Transfer files using FTP commands.

Conclusion

We installed and configured FTP server and client:

- **Server** stores and manages files for sharing.

- **Client** connects to server for uploading/downloading.



13. To configure proxy server.

Theory

- A **Proxy Server** acts as an intermediary between client computers and the internet.
- It forwards client requests to external servers and returns responses back.
- Uses:
 - Improves security by hiding client IPs.
 - Controls access (content filtering).
 - Provides caching for faster response.
 - Monitors network usage.

Steps to Configure Proxy Server

In Windows (Client Side):

1. Go to **Control Panel** → **Internet Options** → **Connections tab**.
2. Click **LAN Settings**.
3. Enable **Use a proxy server for your LAN**.
4. Enter Proxy Server **IP address** and **Port number**.
5. Save settings → Test by browsing.

In Linux (Server Side using Squid):

1. Install Squid package:

Code

```
sudo apt-get install squid
```

2. Edit configuration file `/etc/squid/squid.conf`.

- Set `http_port 3128` (default port).
- Define allowed networks (ACLs).

3. Restart service:

Code

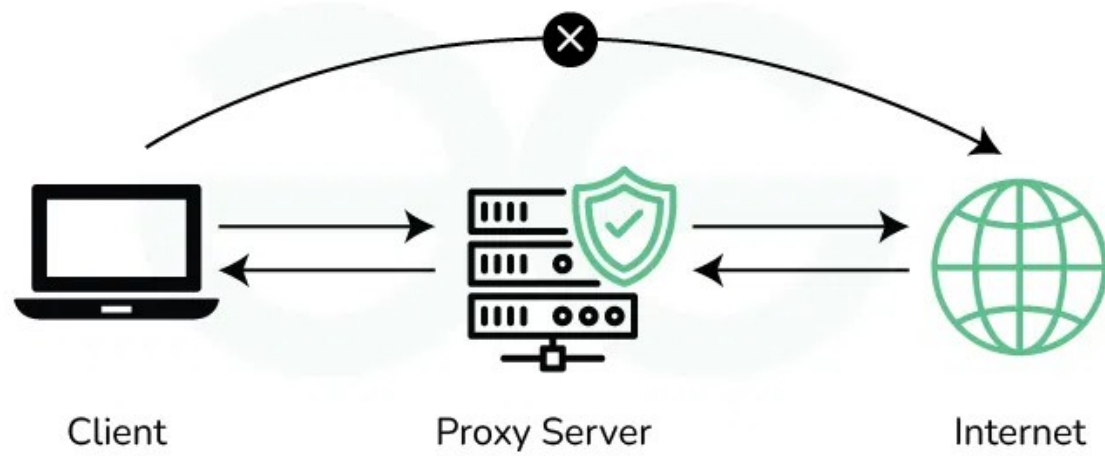
```
sudo systemctl restart squid
```

4. Configure client systems to use server IP and port as proxy.

Conclusion

We configured a proxy server:

- On **Windows**, by setting proxy IP and port in LAN settings.
- On **Linux**, by installing and configuring **Squid proxy server**.
- Proxy servers enhance security, control, and performance in networks.



Proxy Server



14. Familiarization with network simulation tools.

Theory

- **Network Simulation Tools** are software applications used to design, test, and analyze computer networks without needing physical hardware.
- They help in learning, research, and troubleshooting by simulating devices, protocols, and traffic.

Common Tools

1. Cisco Packet Tracer

- Developed by Cisco.
- Used for designing and simulating LANs, WANs.
- Provides virtual routers, switches, PCs, and servers.
- Widely used in networking courses.

2. GNS3 (Graphical Network Simulator 3)

- Open-source tool.
- Allows simulation of complex networks using real router images.
- Suitable for advanced learners and professionals.

3. NS2 / NS3 (Network Simulator)

- Research-oriented simulation tools.
- Used for testing protocols, wireless networks, and performance analysis.
- Provides detailed packet-level simulation.

4. Wireshark

- Network protocol analyzer.
- Captures and analyzes packets in real time.
- Useful for troubleshooting and learning protocol behavior.

Procedure (General Steps)

1. Install the chosen simulation tool (e.g., Packet Tracer).
2. Create a network topology using virtual devices (PCs, switches, routers).
3. Configure IP addresses and protocols.
4. Simulate communication (e.g., using ping).
5. Analyze results and troubleshoot if needed.

Conclusion

We familiarized ourselves with network simulation tools:

- **Packet Tracer** for educational simulations.
- **GNS3** for advanced network emulation.
- **NS2/NS3** for research simulations.
- **Wireshark** for protocol analysis. These tools help design, test, and troubleshoot networks without physical hardware.

